

Figure 5.12: A timeline showing the history of classifying flowering plants into families.
(Source: Royal Botanic Gardens Kew State of World's Plant 2017)



A significant number of major herbaria, including Kew are changing the order of their collections in accordance with APG.

The influential world checklist of selected plant families (also from kew) is being updated to the APG III system.

A recent photographic survey of the plants of USA and Canada is organized according to the APG III system.

In UK, the latest edition of the standard flora of the British Isles written by Stace is based on the APG III system.

5.11 Modern trends in taxonomy

Taxonomists now accept that, the morphological characters alone should not be considered in systematic classification of plants. The complete knowledge of taxonomy is possible with the principles of various disciplines like Cytology, Genetics, Anatomy, Physiology, Geographical Distribution, Embryology, Ecology, Palynology, Phenology, Bio-Chemistry, Numerical Taxonomy and Transplant Experiments. These have been found to be useful in solving some of the taxonomical problems by providing additional characters. It has changed the face of classification from **alpha** (classical) to **omega** (modern kind). Thus the new systematic has evolved into a better taxonomy.

5.11.1 Chemotaxonomy

Proteins, amino acids, nucleic acids, peptides etc. are the most studied chemicals in chemotaxonomy. Chemotaxonomy is the scientific approach of classification of plants on the basis of their biochemical constituents. As proteins are more closely controlled by genes and less subjected to natural selection, it has been used at all hierarchical levels of classification starting from the rank of 'variety' up to the rank of division in plants.

The chemical characters can be divided into three main categories.

1. Easily visible characters like starch grains, silica.
2. Characters detected by chemical tests like phenolics, oil, fats, waxes.
3. Proteins.

Aims of chemotaxonomy

1. To develop taxonomic characters which may improve existing system of plant classification.
2. To improve present day knowledge of phylogeny of plants.

5.11.2 Biosystematics

Biosystematics is an "Experimental, ecological and cytotaxonomy" through which life forms are studied and their relationships are defined. The term biosystematics was introduced by **Camp** and **Gilly** in 1943. Many authors feel Biosystematics is closer to Cytogenetics and Ecology and much importance given not to classification but to evolution.

Aims of biosystematics

The aims of biosystematics are as follows:

1. To delimit the naturally occurring biotic community of plant species.

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